Euler's Equations: Symbolic versus Numerical Mathematics

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Abstract

Three things are to be kept apart: a fluid flow, which is a phenomenon; Euler's equations, which are a model of it; and a proof about the equations, which is a statement about the model. We state the equations, note that they are parameter free, and observe that the mathematical question asked of them — existence, regularity, uniqueness — has not been answered by analysis in symbolic form, because the solutions turn out to have a complicated nature which symbolic analysis has no grip on. Solutions in digital form over a mesh can be computed by a model we name *Adaptive Euler* — a stabilised finite element discretisation together with duality-based output error control and adaptive refinement — and they are turbulent, fluctuating over a range of scales, with well determined mean values such as drag and lift which agree with measurement and arrive with a computed bound on their error. We then ask what a recent symbolic proof of finite-time blowup establishes, and conclude that it is a statement about the model which does not reach either of the other two levels: it exhibits no individual case, so nothing can be inserted into the equations, inspected, or measured.

1 Euler's equations as a model of fluid flow

In 1755 Leonhard Euler presented to the Berlin Academy the memoir *Principes généraux du mouvement des fluides* [4], in which he wrote down the equations of motion of an incompressible fluid. He took them to contain the whole of the mechanics of fluids, and he was explicit that what then stood in the way of using them was not physics but mathematics:

> And if it is not permitted to us to penetrate to a complete knowledge concerning the motion of fluids, it is not to mechanics, or to the insufficiency of the known principles of motion, that we must attribute the cause. It is analysis itself which abandons us here.

What Euler had in mind was *ideal* fluid flow — inviscid incompressible flow — as a model of the slightly viscous, nearly incompressible flow of real fluids such as water and air at subsonic speeds. The equations express Newton's second law and incompressibility, and nothing else.

Let $\Omega \subset \mathbb{R}^3$ be a domain with boundary Γ, let $x = (x_1, x_2, x_3)$ be a Euclidean coordinate in Ω and t time. A fluid of unit density filling Ω is described by a velocity $u(x,t) \in \mathbb{R}^3$ and a pressure $p(x,t)$ satisfying, for $0 < t \leq T$,

$$\dot{u} + (u \cdot \nabla)u + \nabla p = f \qquad \text{in } \Omega, \tag{1}$$

$$\nabla \cdot u = 0 \qquad \text{in } \Omega, \tag{2}$$

together with the *slip* boundary condition

$$u \cdot n = 0 \qquad \text{on } \Gamma, \tag{3}$$

n the outward normal, and an initial velocity $u(\cdot, 0) = u^0$ in Ω.

Equation (1) is Newton's second law. For unit density the mass of a fluid particle is its volume, so force equals acceleration directly: the left-hand side $\dot{u} + (u \cdot \nabla)u$ is the acceleration of a particle following the flow, and the right-hand side is the force acting on it — the pressure gradient $-\nabla p$, which is the force the surrounding fluid exerts, together with any body force f. Equation (2) states incompressibility: the volume of a fluid particle does not change, so the velocity field is divergence free, and the pressure is whatever field enforces this. Condition (3) states that the fluid does not pass through the wall, and nothing more. It is the only boundary condition the equations admit: the normal component is fixed by impermeability, nothing in (1)–(2) determines the tangential velocity at the wall, and prescribing it as well would overdetermine the problem. The no-slip condition belongs to Navier–Stokes, where a second-order viscous term carries the extra condition; here there is no viscosity, and therefore no boundary layer either.

Euler's hope crushed from the start. The expectation that the mechanics of fluids was contained in two short expressions met an objection which was already in print before the expressions were. Three years earlier, in the *Essai d'une nouvelle théorie de la résistance des fluides* of 1752 [6], d'Alembert had shown that a body moving steadily through an ideal fluid experiences *no resistance at all*. Ships meet resistance and cannonballs are slowed, so the conclusion is absurd; and it is a conclusion about exactly the solutions of (1)–(3), to which it duly attached itself. This is d'Alembert's paradox.

It deepened rather than dissolved. In the nineteenth century complex analysis supplied a machinery — the complex potential, conformal mapping — which generated smooth solutions of these equations in profusion, for any body one cared to draw. Every one of them was beautiful and every one gave zero drag. Symbolic mathematics, applied to the equations with its best available tools, produced an ever larger family of solutions none of which described a fluid. Something was wrong and nobody could say what.

It took two and a half centuries to say. The smooth solutions exist, and they are *unstable*: the flow attaches on the forward surface of a body, where it is accelerated, and must separate on the rear surface, where it is retarded, and separation is unstable to spanwise perturbation. The instability generates streamwise vorticity, which stretches and breaks down into a turbulent wake, and the wake carries the drag. The resolution of the paradox on these lines is in [7], and the same mechanism applied to a wing gives lift without a circulation condition imposed by hand [8]. The equations were never wrong. What was wrong was the assumption that the solution of interest is a smooth one.

We note what that resolution required, since it is this paper's subject. It did not come from further analysis of the smooth solutions, of which there were already more than enough. It came from computing the turbulent ones and looking at them.

Setting the viscosity to zero is formal, and the word 'inviscid' misleads. No flow of water or air is inviscid; the viscosity is small, not absent. What the zero stands for is a large

Reynolds number — 10^6 to 10^8 in aerodynamics and ship hydrodynamics — so that viscous forces are negligible beside inertial ones everywhere except in layers which are thin and, as it turns out, do not determine the forces on the body. Euler's equations are the model of that regime, and the regime is where the flows of practical interest live, which is to say most of the fluid mechanics anybody needs. The question is therefore never whether a real fluid has viscosity. It is whether the quantities one wants depend on how much, and we return to it in Section 3, where the computation has a viscosity of its own.

The model is parameter free. There is no constant in (1)–(3) to be measured, fitted or chosen. What enters as data is the *geometry* — the domain Ω, which is to say the shape of the body the fluid flows past — the *forcing* f, and the initial state u^0. Nothing else. Whatever the equations predict is therefore a consequence of the geometry and the forcing alone, and any disagreement with observation is a fault of the model and cannot be repaired by adjustment.

Euler's own verdict, and ours. His position is this paper's, two hundred and seventy years later, with one word changed. The equations were right and remain right. They were not solvable in symbolic form — and in 1755 they were not solvable in numerical form either, there being nothing to compute with. The first of those two impossibilities has not moved. The second has, and the rest of this paper is about what follows from that.

2 Symbolic versus numerical mathematics

The question mathematics asks of a system such as (1)–(3) is about the *properties of its solutions*. Does a solution exist for the given data? Is it regular — does it stay smooth, or do derivatives become unbounded? Is it unique, so that the data determine the flow?

These are the right questions to ask and they have not been answered in symbolic form. The reason is not that the problem has been neglected; it is that the solutions turn out to have a complicated nature. A flow at a Reynolds number of any interest is turbulent: it fluctuates on a range of scales from the size of the body down to scales many orders smaller, with no symmetry, no similarity variable and no closed form. Symbolic analysis works by rewriting, and rewriting needs something to grip. Against an object defined by carrying structure at every scale with no organising principle to exploit, there is nothing for it to take hold of, and this is not a temporary state of the technique but a description of what turbulence is.

What symbolic analysis has produced instead are statements about objects which do possess exploitable structure — self-similar profiles, symmetric configurations, families built by iteration — and conditional bounds of the form: if the velocity is bounded by A, then the derivatives are bounded by an expression exponential in A iterated several times. As theorems these are real. As statements about a fluid they are empty, since a bound of size $\exp\exp\exp A$ is not one any flow can be said to respect. On the symbolic side, in short, analysis still abandons us, exactly where Euler said it did.

What the computer changed. The other impossibility is gone. What was unavailable to Euler was not a method but a machine, and with one the equations can be solved in digital form over a mesh and the solution looked at. What appears when one does is not a formula

and was never going to be: it is a turbulent flow, fluctuating over a range of scales, with a wake, a separation line and a drag — which is to say, the thing Euler's equations were written to describe and which analysis could not reach. The complete knowledge of the motion of fluids he thought barred to us turns out to have been barred only to one of the two ways of doing mathematics.

This is the division the rest of the paper turns on, and it is not a division between rigour and approximation. Symbolic mathematics is exact and reaches objects with exploitable structure. Numerical mathematics carries a tolerance and reaches objects without it. Turbulence is of the second kind, by definition rather than by accident, which is why the computer and not the pen was what the subject was waiting for.

3 Adaptive Euler: solutions in digital form

The same equations treated in digital form behave quite differently. We name the computational model **Adaptive Euler**. It has three parts and the third is not an afterthought:

1. the equations (1)–(3), discretised by a stabilised finite element method over a mesh [5], the stabilisation acting as a vanishing artificial viscosity so that what is computed is a *viscosity solution* of Euler, the limit taken as the mesh is refined;
2. *duality-based output error control*: for each output of interest — a drag, a lift — the error is bounded by the residual of the computed solution weighted by the solution of an adjoint problem, which measures how much the residual matters for that output;
3. *adaptivity*: the mesh is refined where the weighted residual says the error lives, and not uniformly, until the bound on the output is below what is asked.

The error control is part of the model rather than an assessment of it. What Adaptive Euler returns is not a flow but a *pair*: an output and a bound on how wrong that output may be, computed together. A model which returned the first without the second would be a different model and a much weaker one, since nothing could then be concluded from it.

The viscosity introduced, and why its size does not matter. A computation at finite resolution cannot have zero dissipation, so Adaptive Euler necessarily introduces a viscosity. This is the obvious objection to it — that what is being solved is not Euler's equations but some nearby viscous system — and the answer is a computational fact rather than an argument.

The mean-value outputs do not depend on the absolute size of the viscosity introduced. Refine the mesh: the artificial viscosity falls, the resolved scales multiply, the flow field changes in every detail — and the drag does not move. The dissipation rate does not fall with the viscosity either; it settles at a value fixed by the geometry and the free stream. That independence is the dissipation anomaly in computational form, the phenomenon Onsager accounted for [13] and Taylor measured, and it is the reason the limit is meaningful at all.

It also places the computation and the fluid in the *same* position rather than in different ones. Air has a small viscosity whose absolute size does not determine the drag of a wing; Adaptive Euler has a small viscosity whose absolute size does not determine the computed

drag. Neither is inviscid and neither needs to be. What both exhibit is a flow in which the viscosity sets the smallest scale present and nothing that is asked about.

What Adaptive Euler produces has the following properties, and none of them was put in.

It is turbulent. The computed flow fluctuates, in time and across a range of scales widening as the mesh is refined. It is not a steady solution, not a symmetric one, and not one that could have been guessed.

Its mean values are determined. The flow field itself is not reproducible in detail — and could not be, since for weak solutions of Euler uniqueness is false. But mean-value outputs are: drag, lift, the position of the separation line. These converge under refinement, and each arrives with the bound of part 2 above, so the answer carries a statement of how wrong it may be.

They agree with measurement. Drag within a few per cent, the separation line where experiment puts it, the lift curve and the stall, the drag crisis [9]. These are quantities measured by other people, and the comparison is not one we administer: the cases, the geometries and the measurements belong to the benchmark workshops, and predictions are submitted before the measurements are released. Adaptive Euler has been put through them — the high-order CFD workshop case CS1 [10], and validation against the NASA High Lift Prediction Workshop 4 [11], both by Jansson — which is the form of accountability a computational model can have and an existence proof cannot.

And it is a solution of Euler. That the limit satisfies (1)–(3) weakly is not an identification offered because the numbers look right. The stabilisation of the method is a weighted least-squares control of the residual; the discrete energy estimate bounds that residual in a weighted norm; testing against a smooth function sends it to zero in the sense the weak formulation requires. The limit is a weak solution, as a theorem.

So the questions of Section 2 do have answers in digital form, though not the answers the symbolic form was looking for. Existence: by construction and exhibition. Regularity: none — the velocity stays bounded while its gradients do not, the solution being Hölder continuous with exponent about $1/3$ — the exponent below which Onsager observed that energy conservation can fail [13] — and dissipating energy at a rate independent of the vanishing viscosity. Uniqueness: false for the flow field, true for the mean values, which is the form in which the question has an affirmative answer and the form in which anybody wants one.

No AI is needed for Euler. The equations are parameter free, and the parameters of an AI have no role to play.

That is the whole of it. The equations are given, in symbolic form, in three lines; there is nothing to infer from data because Newton's second law and incompressibility are not hypotheses to be fitted. The discretisation is derived from the equations, the error estimate from the residual and the adjoint, the adaptivity from the estimate. No quantity enters anywhere that was obtained by fitting to observations. The model carries no parameters, Adaptive Euler carries none either — a mesh and a tolerance are choices stated by the user, not constants

calibrated against the answer wanted — and a large language model carries of the order of 10^{11} to 10^{12}, obtained by optimisation against a corpus.

Trained closures and surrogates exist and are pursued, and we make no claim about them in general. We observe only that any such component reintroduces fitted parameters, and that this is the property Section 1 found the model to be free of. A parameter-free prediction which disagrees with measurement condemns the model. A prediction carrying a trained component which disagrees invites retraining. The first is a scientific statement and the second is a procedure.

4 Inspection of blow-up proofs by OpenAI

Against this stands a recent proof, produced by an AI system and formalised in Lean, that there exists smooth, compactly supported, divergence-free initial data on $\mathbb{R}^3$ whose smooth Euler solution has finite maximal lifespan: the velocity gradient becomes unbounded, and the Beale–Kato–Majda [12] vorticity integral diverges [1]. Comparable results were being prepared independently and by hand [2]. We take the proof to be correct, and nothing below depends on whether it is.

It is a statement about the *model*, and it is a real one: a question open for decades is settled. Three observations place it.

What blows up is the gradient, and that is what turbulence does too. The word blowup invites the reading that the velocity becomes infinite. It does not, and the theorem does not claim it. What is asserted is

$$\limsup_{t\uparrow T_*} \|\nabla u(t)\|_{L^\infty} = \infty, \qquad \int_0^{T_*} \|\operatorname{curl} u(t)\|_{L^\infty}\, dt = \infty,$$

the gradient and the vorticity integral, with nothing said about the velocity becoming unbounded.

That is precisely the signature of the computed turbulent solution of Section 3: gradients diverging while the velocity stays bounded. The two accounts therefore agree on the fact, that smooth solutions of (1)–(3) break down in finite time with unbounded gradients, which is what we reported computationally in [3]. The agreement is larger than the vocabulary of blowup against turbulence suggests, and it is worth stating before the differences.

But the same quantity diverges for opposite reasons. In the turbulent flow the gradient grows because the cascade carries energy to smaller and smaller scales and *the energy leaves*; the divergence of ∇u is the mechanism of dissipation, which is the drag. In the construction the gradient grows because the energy *cannot* leave — the solution is classical and conserves it exactly — so a growing gradient can be accommodated only by concentrating what is there. One divergence is energy departing, the other is energy piling up. The symptom is shared and the physiology is opposite.

They differ also in kind of event. In the turbulent solution the unboundedness of the gradient is a *persistent condition*: the solution goes on existing, as a weak solution, dissipating,

for as long as one cares to run it. In the construction it is a *terminal event*: at T_* the smooth solution ceases, and the theorem says nothing about what follows.

Which locates the result rather exactly. T_* is the end of the laminar regime. That is a real thing to have established, and we would not have it dismissed: the breakdown of smooth Euler flow in finite time was asserted on computational evidence and is now proved, for one constructed datum. But a fluid spends essentially none of its life before T_* and essentially all of it after. The flows of Section 3 — a wake, a separation line, a drag — are what happens beyond the breakdown, and they are what the equations were written to describe. The theorem stops at the point where fluid mechanics begins.

It conserves energy, and therefore cannot be a fluid. With smooth data the solution is classical up to the blowup time, and a classical solution of (1)–(2) with decaying data satisfies

$$\frac{d}{dt}\int \tfrac{1}{2}|u|^2 = 0,$$

by an integration by parts. The construction therefore dissipates *nothing* over the whole of its existence. The singularity is not a mechanism by which energy is lost but one by which it is piled up: at the singular time all of it is still present, gathered into a vanishing region. A fluid loses energy continuously from the moment it meets a body, and the loss is the drag.

It exhibits no individual case. The datum is the limit of an iteration over stages and has no closed form. Nothing can be plotted, nothing measured. Nor can either check on a *result* be applied: there is no symbolic expression to insert into (1)–(2) and reduce, and no numerical field either, since no finite stage of the iteration blows up — the property belongs to the limit. What exists is the existence.

What is proved was already claimed. There is a further observation, and it is not ungenerous. The theorem asserts that *some* smooth datum has finite lifespan. Our position, stated in [3] in 2008 and resting on computation, is stronger: *every* laminar Euler flow at high Reynolds number is unstable and breaks down, with the velocity staying bounded while the gradients do not. The existence statement follows from ours as a corollary; if all smooth solutions break down then certainly one does.

What is new is therefore the proof and not the fact. That is a genuine contribution — a fact asserted on computational evidence is not the same as a fact established — and we do not diminish it. But it should be seen for what it is: a demonstration, in a special constructed case, of a phenomenon which is generic, which has been exhibited computationally for two decades, and which the instability of Section 2 predicts of every flow past every body.

And the viscous case does not say what it is taken to say. Two things about the companion proof for Navier–Stokes, offered without disputing it.

The first is a matter of what was proved. Fefferman's four alternatives do not form two pairs of negations: (A) and (B) are posed with $f \equiv 0$ and quantify over *all* admissible data, while (C) and (D) ask for the existence of admissible data *together with an admissible smooth*

forcing for which smoothness fails. A proof of (C) therefore leaves (A) untouched. It does not show that a fluid tears itself into a singularity; it shows that *something can be done to a fluid* which does. Those are different statements and only the first would be about fluid flow.

The second is a matter of scale. Navier–Stokes rescales: if v solves the equations at unit viscosity with forcing g, then

$$u(x,t) = a\,v(bx,ct), \qquad a=\nu b, \quad c=ab, \quad f=a^2 b\,g, \tag{4}$$

solves them at viscosity ν, and the map is onto — one unit-viscosity construction generates the whole family. Running it down to the viscosity of a fluid then forces a choice, and there are only two. Hold the solution at size one, $a=1$: the forcing amplitude is $a^2b=\nu^{-1}$, so the force required diverges as the viscosity falls and leaves any fixed admissible class. Or hold the forcing, $a^2b=1$: then

$$\text{amplitude } \nu^{1/3}, \qquad \text{length } \nu^{2/3}, \qquad \text{time } \nu^{1/3},$$

all tending to zero — at $\nu=10^{-6}$ the velocity reaches 10^{-2} on a scale of 10^{-4} before blowing up.

Set that beside the flow the problem is about. Turbulence at small viscosity has amplitude *fixed* and structure refining. The constructed family has amplitude *vanishing* like $\nu^{1/3}$, which is the opposite behaviour, or else a forcing diverging like ν^{-1}, which is not a fluid at all. Neither branch survives the limit and there is no third. An independent exclusion of the mechanism under analytic forcing points the same way [15].

So the construction requires either a force no fluid is subject to, or a singularity too small to constitute anything, and Horace supplies the description: *parturient montes, nascetur ridiculus mus.*

Forwards and backwards. The two constructions of this paper are of the same formal kind and opposite in direction, and it is worth saying so plainly. The singular solution fixes the singularity wanted and solves *backwards* for data and a force that will produce it, the force being whatever the residual requires. Adaptive Euler fixes the data a flow actually has and solves *forwards*, the residual becoming the bound on what may be asked of the result. Both are automated and both exhibit something. Only the second exhibits a fluid.

5 Three levels, and what reaches which

The title names three things and the point of the paper is that they are three.

A fluid flow is a phenomenon. It has a drag which can be measured.

Euler's equations are a model of it, parameter free, with geometry and forcing as data. They are answerable to the phenomenon: if the drag they give is wrong, nothing can be adjusted to repair it.

A proof about the equations is a statement about the model. It is answerable to the model alone.

The digital solution reaches all three levels. It is a solution of the equations, by the residual control; it is an object that can be inspected, and inspecting such objects is where the resolution

of d'Alembert's paradox [7] and the account of flight [8] came from; and its mean values can be set beside a measurement, where they survive.

The symbolic proof reaches the first of the three and stops. That is not a criticism of it as mathematics, where it is an advance, and existence without an exhibited instance is ordinary and accepted practice — the disproof of the Mertens conjecture [16] established that a counterexample exists and exhibited none. It is a statement about what the result can be offered as. A theorem about a model becomes a statement about fluids only by way of an instance, and there is none here.

6 The shock, and the question it raises

The OpenAI proofs of singularity formation for Euler and for Navier–Stokes have sent a shock wave through the mathematical community, and the question being asked is plain enough. If a machine can produce proofs of theorems better than people can, what is left for those whose main occupation is proving theorems?

The question is legitimate and we do not propose to soften it. But two things about the present case bear on it, and both follow from the argument above.

What was automated was the derivation, not the judgement. The system worked on a problem already posed, already named, already agreed to be worth settling, and already being attacked by others along related lines [2]. It did not decide that finite-time blowup for Euler was the question to ask. It did not decide that Euler's equations were the right model of a fluid, which is a judgement about the world and not about symbols. And it did not connect its theorem to anything: the step from a statement about a model to a statement about fluids is not a proof step, and nothing in the output performs it. Sections 4 and 5 are an instance of that gap rather than a complaint about it.

And this has happened before, to the discipline in this paper. Machines outran hand calculation in the 1950s, and numerical analysis did not end. It changed what it was about. The object of study moved from finding a solution — which the machine would do — to bounding the error in what the machine produced, to deciding what was worth computing, and to establishing when a computed quantity is determined by the equations at all and when it is not. That is a harder subject than the one it replaced, and it is the subject that produced the error control of Section 3.

We would therefore expect the answer for mathematics to have the same shape, without pretending to know it. Deriving will be automated, and probably well. What will not be is choosing the question, judging whether a model deserves the attention, supplying the instance which lets a theorem touch something, and saying what a result means. On the evidence of this episode those are separate activities from derivation, because here they were left undone.

And there is relevant symbolic mathematics, which is not theorem-proving. If the proofs discussed here concern the model and not fluids, that is no reflection on symbolic mathematics as such. It is a reflection on one use of it. The most consequential symbolic work in physics has never been the proving of theorems about given equations. It has been the *formulation of the equations*, from principles.

Equations (1)–(3) are themselves such a piece of work. Newton's second law is a statement

about particles; writing it for a continuum, deciding that the force of the surrounding fluid is a pressure gradient, and adjoining incompressibility as a constraint with the pressure as its multiplier, is symbolic mathematics of the first order — and every step of it required deciding what a fluid *is*. The result is three lines carrying no parameters, from which everything in this paper follows. Schrödinger's equation is the same kind of achievement for atoms, and the reformulation of atomic physics as non-overlapping charge densities meeting at free interfaces, which one of us has pursued under the name RealQM [14], is an attempt at the same activity rather than at theorem-proving: the work is in what the equations should be, not in what follows from them.

This is where symbolic mathematics does its heaviest lifting, where physical judgement enters mathematics, and where, so far, no machine has done anything at all. A system which proves a theorem about Euler's equations has been handed the hard part.

7 Summary

The equations (1)–(3) are parameter free and have only geometry and forcing as data. The mathematical questions asked of them — existence, regularity, uniqueness — have gone unanswered in symbolic form because turbulent solutions have no structure for symbolic analysis to exploit, this being what turbulence is rather than a gap in technique. In digital form, by Adaptive Euler, the same questions are answered, in altered shape: solutions exist by exhibition, are not regular, are not unique as fields, and are unique in the mean values anybody wants, with computed error bounds and agreement with measurement.

A symbolic proof of finite-time blowup settles a long-open question about the model. Its solution conserves energy exactly and so dissipates nothing, whereas a fluid dissipates and the dissipation is the drag; and it exhibits no individual case, so nothing can be inserted into the equations, inspected or measured. It is an exact statement about an object with no observables, set beside approximate statements about observables which agree with experiment. Each has what the other lacks, and only one of them is answerable to a fluid.

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